

Oscar (Yan) Yu

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EDUCATION

Honors Computer Science, Honors Business Administration

October 2025

Western University, Ivey Business School, HKUST (Exchange)

London, ON

- **Scholarships/Awards:** Western Scholarship of Distinction, Dean's Honour List (2020-2025)
1st place Western University team @ International Collegiate Programming Contest (ICPC) NA Qualifiers (2021, 2022)
- **Relevant Courses:** Data Structures and Algorithms, Distributed Systems, Cybersecurity Management, Operating Systems, Databases, Computer Networks, Computer Architecture, Machine Learning, Human Computer Interaction and Design

SKILLS

Languages: Python, Java, C++, C, Typescript/Javascript, C#, SQL (MySQL, Postgres, SQLite, Oracle DB), HTML/CSS
Frameworks/Libraries: Node.js, Next.js, React, Flask, FastAPI, Qt, Spring Boot, gRPC, Pandas, NumPy, TensorFlow, Keras
Tools: Git, Unix/Linux, Docker, Azure, AWS, Apache Spark, Databricks, Redis, MongoDB, GraphQL

EXPERIENCE

Mark43

June 2024 – Aug. 2024

Software Engineering Intern

New York, NY

- Developed updated backend data collection and validation modules, database schemas, and data classes using Java and MySQL to enable custom configurations for new metadata-based document querying and transformation features.
- Implemented new API endpoints, UI components, and date logic for client-configurable reporting periods using TypeScript and React, enabling federal and state level NIBRS compliant reporting for tenants in previously unsupported regions.
- Optimized CI/CD pipelines using Github Actions and TeamCity to eliminate deprecated tests and decouple cross-repository microservice dependencies, reducing testing-related downtime for deployments by more than 40 minutes.

CIBC

May 2023 – Aug. 2023

Software Developer Intern

Toronto, ON

- Architected department-wide data archival system and implemented data pipelines using Python and Azure Data Factory to automate data compression and archival, reducing live archive size and hosting costs by 80%.
- Optimized global config management using Azure Databricks to consolidate configuration files into a single source of truth, eliminating version mismatches and reducing pipeline development/maintenance time by up to 90%.
- Modularized C# API templates, enabling discretionary querying and eliminating redundant API calls for business teams.

Bank of Montreal

May 2022 – Aug. 2022

Risk Analyst Intern

Toronto, ON

- Created dashboards using Python and PowerBI, enabling detailed counterparty risk tracking for 50+ market risk analysts.
- Automated daily FX report generation from Oracle DB via Python and Chron scripts, reducing daily data latency by 90%.
- Built data pipelines between database systems and internal reporting/analytics using Python to streamline complex SQL querying across 10+ tables and 6 million rows, decreasing time spent on manual querying by 50%.

LEADERSHIP

Hack Western, Web Developer — Next.js, React, TypeScript, tRPC, SQL

Apr. 2024 – Dec. 2024

- Designed custom application review system and tRPC APIs, reducing average organizer review flow to under two minutes.
- Deployed PostgreSQL and Drizzle ORM database schemas on DigitalOcean for hacker applications and internal tooling.
- Implemented internal organizer dashboards, hacker application pages, and sponsor showcases using React and TypeScript.

Western AI, Project Manager — Python, TensorFlow, Keras, Streamlit

Sept. 2022 – Apr. 2023

- Led weekly meetings and mentored team for 5 to develop a convolutional neural network using Python and TensorFlow.
- Deployed 12-million parameter CNN and tuned hyperparameters to classify 500+ bird species with 85% accuracy.

PROJECTS

Western Rank 🏆🏆 — Next.js, TypeScript, Python, SQL

- Built a course review platform for over 30 000+ students @ Western University and affiliate colleges.
- Designed Next.js REST API endpoints for reviews and course data hosted on Amazon RDS (Postgres).
- Automated description and prerequisite linking of 6000+ courses using Python, BeautifulSoup, and Pandas.

Game Boy ++ 🎮 — C++, SDL2

- Emulated the Nintendo Game Boy architecture in C++ for Raspberry Pi by reverse-engineering hardware documentation.
- Implemented complete CPU instruction set, registers, memory management system, and graphics emulation using SDL2.
- Increased code coverage to 80% by writing unit tests using Catch2 and test ROMs to validate Game Boy functionality.

CPR Redis 🗄️ — Python, gRPC, Redis

- Designed a low-latency, consistent, distributed in-memory database solution for high performance caching applications.
- Implemented the RAFT election algorithm using Python and gRPC calls to synchronize independent Redis instances.
- Developed JSON-based persistent logging solution to restore/recover Redis instances from scratch on launch or restart.